GUVI × HCL CAPSTONE · RECOMMENDATION SYSTEMS

Audible Insights: Intelligent Book Recommendations

A book recommendation system for 6,056 Audible India titles: data cleaning and merging, NLP on descriptions, clustering, and content-based, clustering-based and hybrid recommenders, served through a Streamlit app on Streamlit Community Cloud and AWS.

Author	Aryaman Bhatnagar
Date	18 October 2026
GitHub	https://github.com/aryaman2213/audible-insights
Stack	Python, pandas, scikit-learn, SciPy, Plotly, Streamlit, Docker, AWS EC2

6,056
unique books from 10,832 raw rows

16 clusters
silhouette 0.339; DBSCAN agrees (ARI 0.98)

0.218
hybrid precision@10, genre-blind (5.2× random)

0.251
rating RMSE vs 0.274 for the mean

1. Problem statement and business use cases

Design a book recommendation system that cleans and merges two Audible datasets, applies NLP and clustering, builds several recommendation models, compares them with evaluation metrics, and serves recommendations through a user-friendly Streamlit interface deployed on the cloud.

- **Personalised reading.** "Similar books" explains every pick; "For you" works from genres, sub-genres, authors or liked books, with no history needed.
- **Libraries and bookstores.** Top-rated, most-reviewed and hidden-gem shelves to drive borrowing and sales.
- **Authors and publishers.** Which genres have the most titles versus the most listeners, how ratings behave, and how pricing tracks length.
- **Reader engagement.** Trending and highly rated books across 17 genres.

2. Data preparation and cleaning

`Audible_Catlog.csv` (6,368 rows: name, author, rating, reviews, price) and `Audible_Catlog_Advanced_Features.csv` (4,464 rows: the same plus description, listening time, ranks and genre). Note that the brief describes these the other way round: the "advanced" file is the one with descriptions and genres.

Problem found	Count	Action
Exact duplicate rows	929 + 234	Dropped
Same book scraped twice with different values	266	Kept the most complete copy, then the most reviewed
Author spelled two ways for the same title	3	Fuzzy match (≥ 0.85) before merging
Rating –1 (always with missing reviews)	631 + 421	Marked unrated, never averaged in
Descriptions that are scraping failures: CAPTCHA pages, Amazon error pages, checkout text, publisher stubs, award one-liners	2,022	Treated as missing
Ranks & Genre / listening time missing	2,268 / 2,271	Left missing; genre inferred by NLP where confident
Prices 10× too high (dropped decimal)	132	Divided by 10 and flagged (same length as other books, 14× the price per hour)
876 fine genres in one rank string	n/a	Parsed (split on `,#`), mapped to 17 broad genres by ordered keyword rules

Merge: outer join on normalised title + author. The advanced (later) scrape wins where both files have a value. Result: 6,056 books (3,347 in both files, 2,049 only in the basic one, 660 only in the advanced one). Coverage after cleaning: 5,479 rated, 2,298 with a real description, 2,059 with a listed genre.

Derived: Bayesian weighted rating (m = 50 reviews), popularity (log reviews), hidden-gem flag (rating ≥ 4.5, 5–65 reviews), listening minutes, overall Audible rank.

3. Exploratory data analysis

Distributions and outliers after cleaning

Field	Lower fence	Upper fence	IQR outliers	Share	Action
Rating	3.7	5.3	133	2.4%	kept: low ratings are real (mostly books with few reviews)
Reviews	-908.5	1607.5	741	12.2%	kept: real blockbusters; log-scaled (popularity) where used
Price (₹)	-79.5	1468.5	141	2.3%	10× decimal-shift errors already fixed; the rest are genuinely long or premium titles
Listening (min)	-359.8	1406.2	126	6.0%	kept: genuine long books and collections

Ratings barely vary (std 0.35, skew -3.1). Review counts are extremely skewed (median 190, max 70,122), hence log(reviews) as *popularity* and the Bayesian weighted rating.

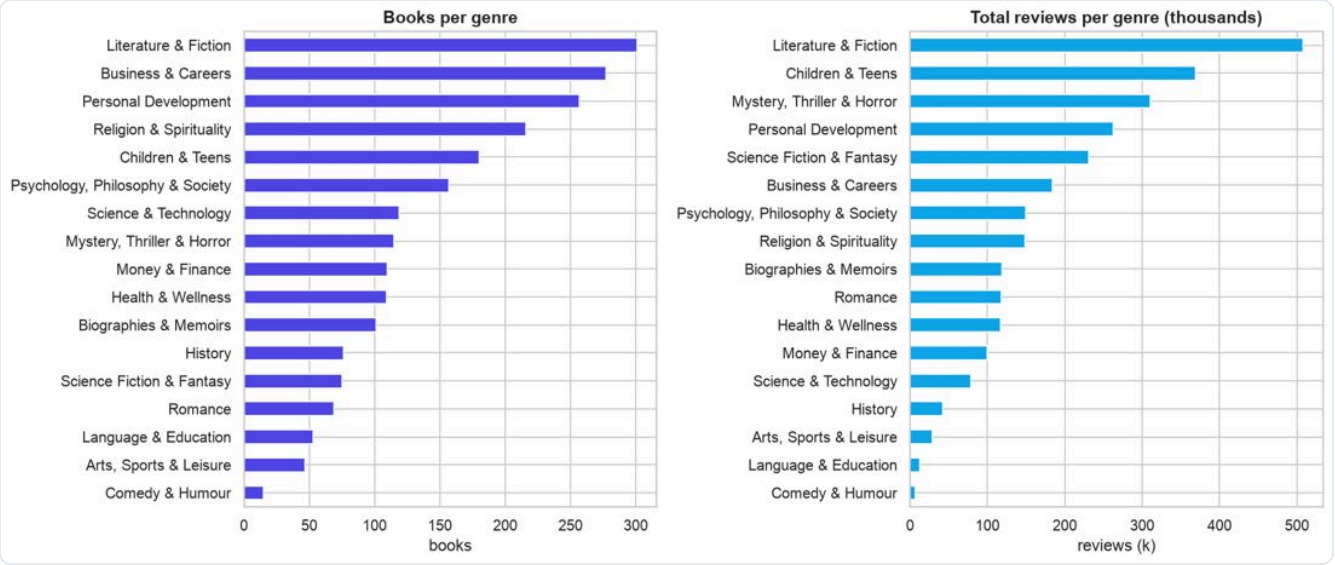
Q1. What are the most popular genres in the dataset?

It depends on what "popular" means:

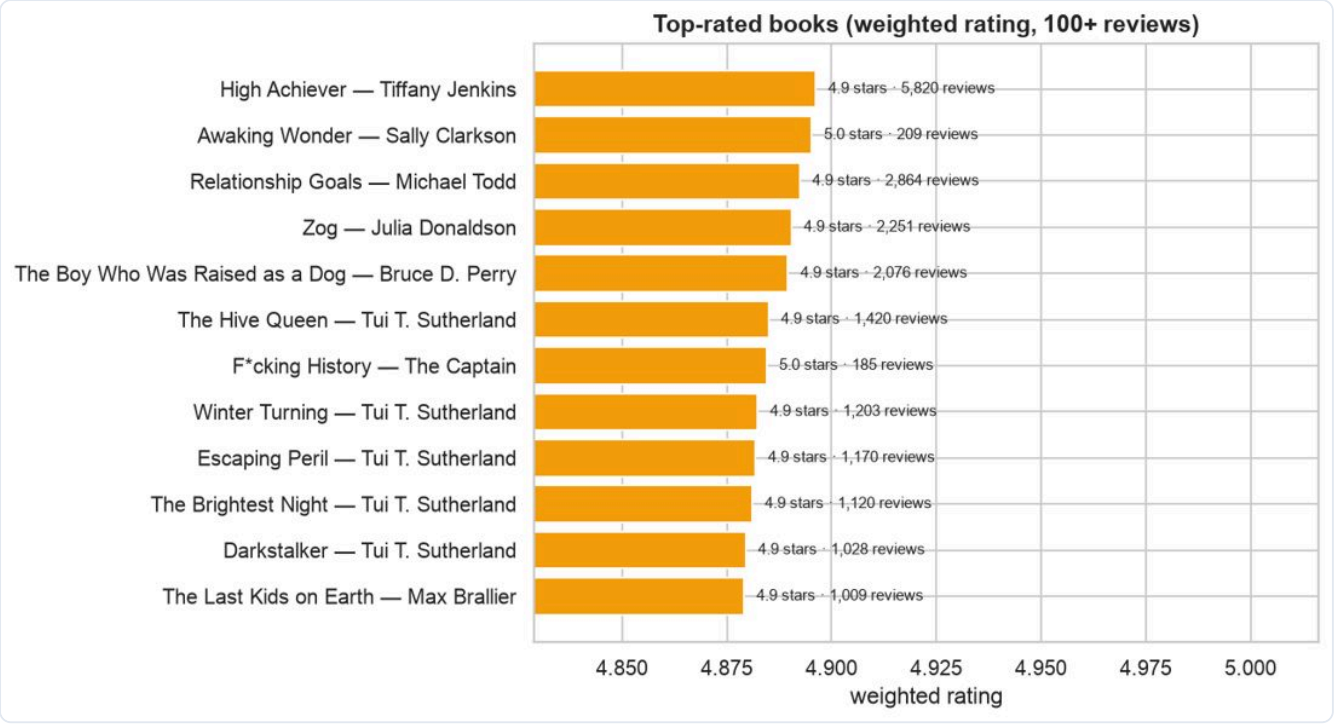
- **By number of titles:** Literature & Fiction (301 books), Business & Careers (277) and Personal Development (257) lead. The catalogue leans heavily towards non-fiction self-improvement.
- **By listeners (total reviews):** Literature & Fiction (18% of all reviews), **Children & Teens (13%)** and **Mystery, Thriller & Horror (11%)**. Business is only 6% of reviews despite being the second-largest shelf.
- **By engagement per title:** Science Fiction & Fantasy has the highest median reviews per book (1,412), about 8× Business (183). Thrillers come second (889). These genres have fewer titles but much bigger audiences per title.

The most common *fine* genres are Personal Success, Personal Development & Self-Help and Analysis & Strategy.

Top-rated books (weighted rating, at least 100 reviews) are led by *High Achiever* (4.9, 5.8k reviews), *Awaking Wonder* (5.0), *Relationship Goals* (4.9, 2.9k) and Julia Donaldson's *Zog* (4.9, 2.3k). Tui T. Sutherland's *Wings of Fire* series takes six of the top 15 places.



Q1. Books and total reviews per genre



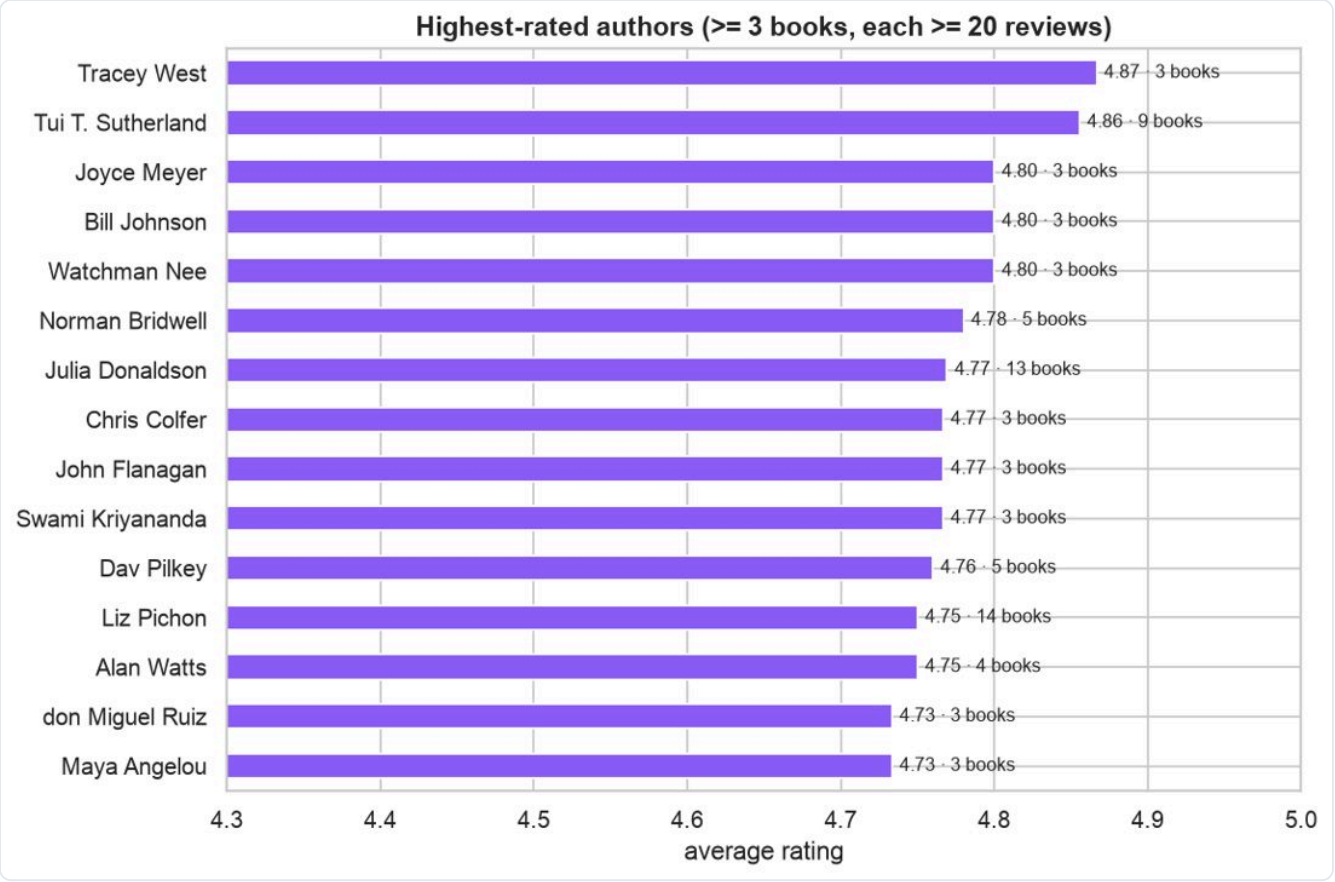
Top-rated books (weighted rating, 100+ reviews)

Q2. Which authors have the highest-rated books?

I only counted authors with at least 3 rated books, each with at least 20 reviews, so one lucky 5-star book can't top the list. **Tracey West (4.87)**, **Tui T. Sutherland (4.86 over 9 books, 10k reviews)**, Joyce Meyer, Bill Johnson and Watchman Nee (4.80) lead. Julia Donaldson (13 books, 30k reviews, 4.77) is the most remarkable: consistently top-rated at scale.

Children's authors (Sutherland, Bridwell, Donaldson, Colfer, Pilkey, Pichon) and Christian/spiritual writers dominate the top. Both genres rate above the catalogue average (4.59 and 4.56 vs 4.46).

Most common authors (by number of titles): Harvard Business Review and OSHO (32 each), Devdutt Pattanaik (29), Ruskin Bond and Innovative Language Learning (28), Rick Riordan (26) and Agatha Christie (24). Being prolific is not the same as being loved: Innovative Language Learning averages 3.87, while Rick Riordan averages 4.69 across 67k reviews.



Q2. Highest-rated authors

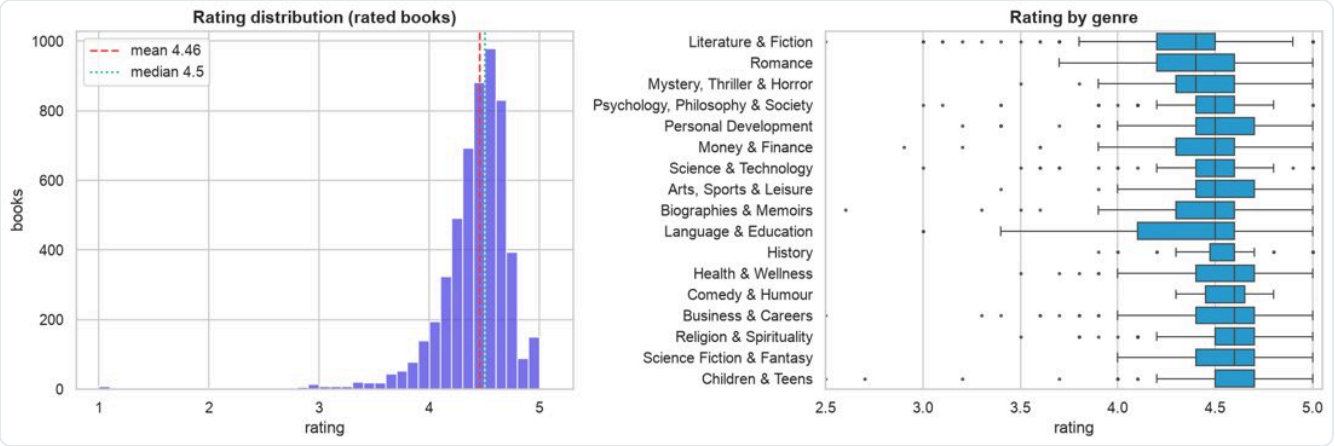
Most common authors (by number of titles):

Author	Titles	Avg rating	Total reviews
Harvard Business Review	32	4.46	2,821
OSHO	32	4.48	192
Devdutt Pattanaik	29	4.27	7,374
Ruskin Bond	28	4.51	6,773
Innovative Language Learning	28	3.87	1,827
Rick Riordan	26	4.68	67,515
Agatha Christie	24	4.51	6,754
Brian Tracy	22	4.47	5,183

Q3. What is the average rating distribution across books?

Ratings are crowded at the top. The mean is **4.46**, the median **4.5**, and the standard deviation only 0.35. **61% of rated books score 4.5 or more**, and only 5.6% fall below 4.0 (skew –3.1). A 4.3 is therefore below average here, which is why the recommenders use a *weighted* rating (shrunk towards the mean when reviews are few) rather than the raw stars.

By genre, Children & Teens, Religion & Spirituality, Comedy and Science Fiction & Fantasy rate highest. Literature & Fiction (4.32) and Language & Education (4.34) rate lowest.

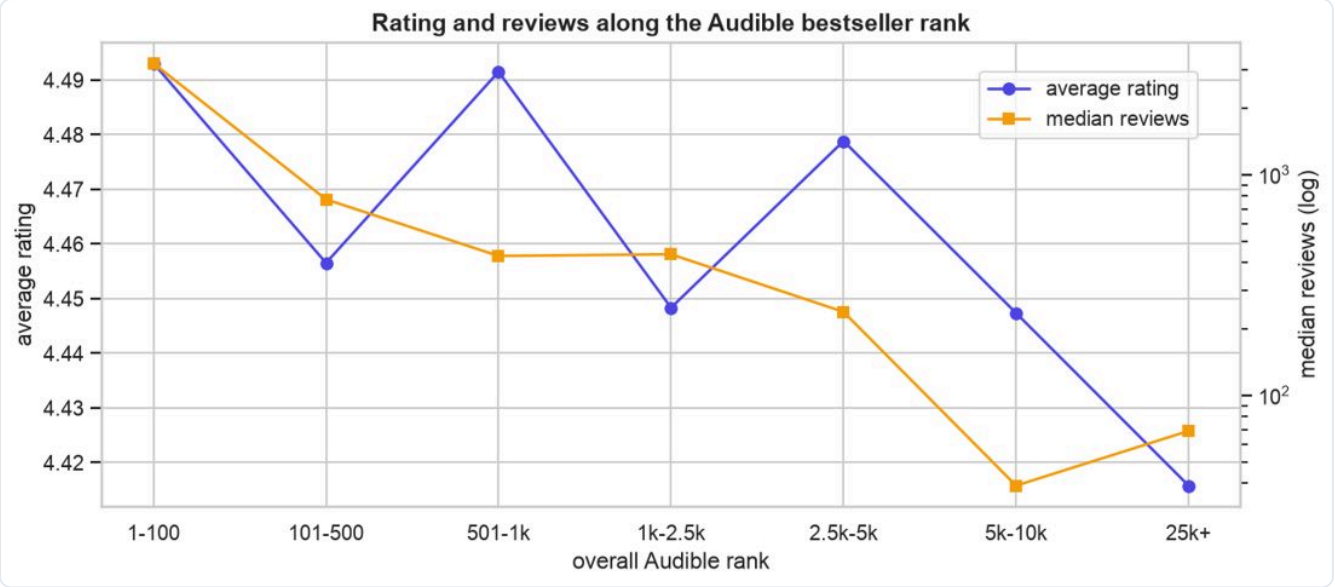


Q3. Rating distribution and rating by genre

Q4. Are there trends in publication years for popular books?

Neither CSV has a publication date. Only 17 descriptions state one ("first published in 1936") and 40 titles contain a year, which is far too few for a trend.

What the data *does* have is the overall Audible bestseller rank, so the line chart above is the closest honest substitute. Moving down the rank, median reviews fall steeply (3,197 for the top 100 → 239 at rank 2.5k–5k), while average rating barely moves (4.42–4.49). **Popularity and quality are almost independent in this catalogue.** To answer the original question, publication years would have to be joined in from an external source such as the Open Library or Google Books API (matched on title + author). That is listed as future work.



Q4 substitute: rating and reviews along the Audible bestseller rank (no publication dates exist)

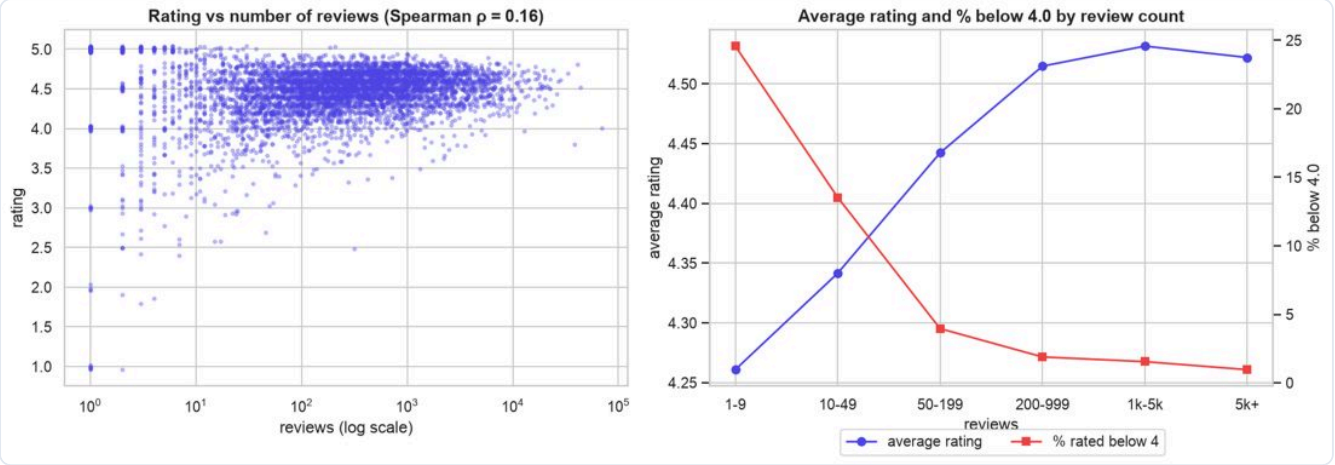
Q5. How do ratings vary between books with different review counts?

Books with few reviews have far more extreme ratings, in both directions:

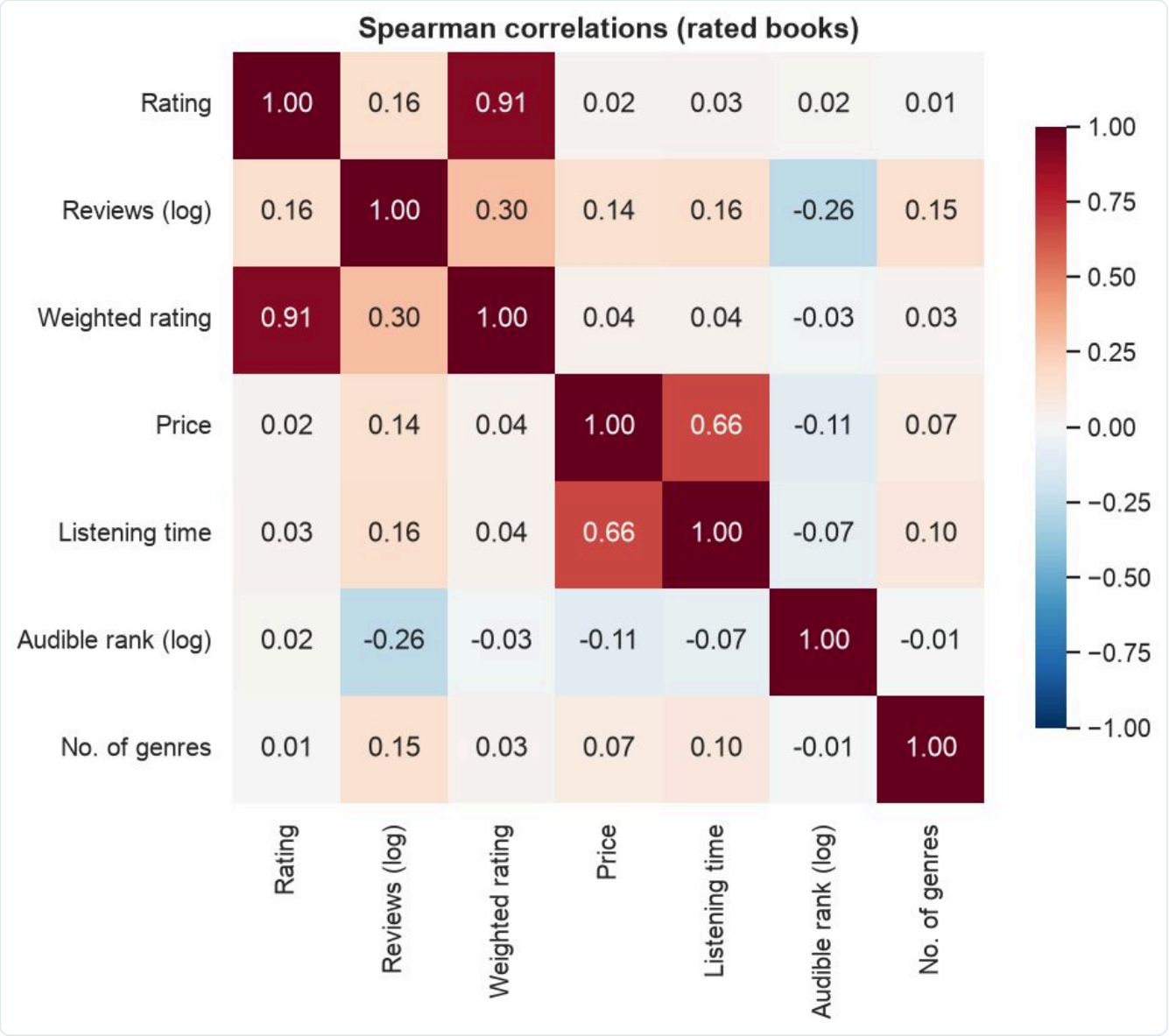
- With 1–9 reviews, **33% are perfect 5.0s and 25% are below 4.0** (std 0.86).
- From 200 reviews up, almost nothing is 5.0 or below 4.0 (std ≈ 0.2).

The average rises with review count, from 4.26 to about 4.52, then flattens after 1,000 reviews. The overall correlation is weak but real (Spearman $\rho = 0.16$, $p < 10^{-32}$).

Two practical consequences. A raw 5.0 is mostly a small-sample artefact. And widely read books are rarely bad. This is exactly why the system ranks by Bayesian weighted rating (m = 50 reviews).



Q5. Ratings vs number of reviews



Correlations between numeric features

Statistical tests

All non-parametric, because ratings are bounded, discrete and heavily skewed.

Question	Test	Why this test	Statistic	p-value	Result
Do ratings differ across the 17 genres?	Kruskal–Wallis H	compares >2 groups without assuming normality	218.768	1.2e-37	yes: genre matters
Are books with 1,000+ reviews rated differently from books with <50?	Mann–Whitney U	two independent, skewed samples	510,803.500	1.3e-19	yes: median 4.6 vs 4.4, and far less spread
Is rating related to review count?	Spearman ρ	monotonic relation, robust to outliers	0.162	1.4e-33	weak positive (ρ = 0.16)
Does the share of free titles depend on genre?	Chi-square independence	two categorical variables	173.328	1.8e-28	yes: free titles concentrate in Language & Education

4. NLP and clustering

Text features

Title (twice) + cleaned description → TF-IDF (1–2-grams, sublinear tf; English, audiobook and publisher stop words such as "Penguin presents" and "Audie award").

> **Genre inference.** Logistic regression on TF-IDF, trained on 1,821 books with known genres (17 classes). 5-fold CV accuracy 52.0%, macro-F1 0.479, top-3 accuracy 75.8%. The confidence cut-off (0.17) is chosen on CV so the predictions kept are right 70% of the time. That filled in 218 of 477 books that had a description but no genre; the rest stay "Unknown".

Clustering

The text and genre blocks are reduced with TruncatedSVD to 150 topics. Clusters are learned on the 2,059 books with listed genres (KMeans, k chosen by silhouette over 8–30). Every other book then joins the nearest cluster centroid in a text-only space. Clustering everything together left the genre-less books in one leftover cluster.

Q6. Which books are frequently clustered together based on descriptions?

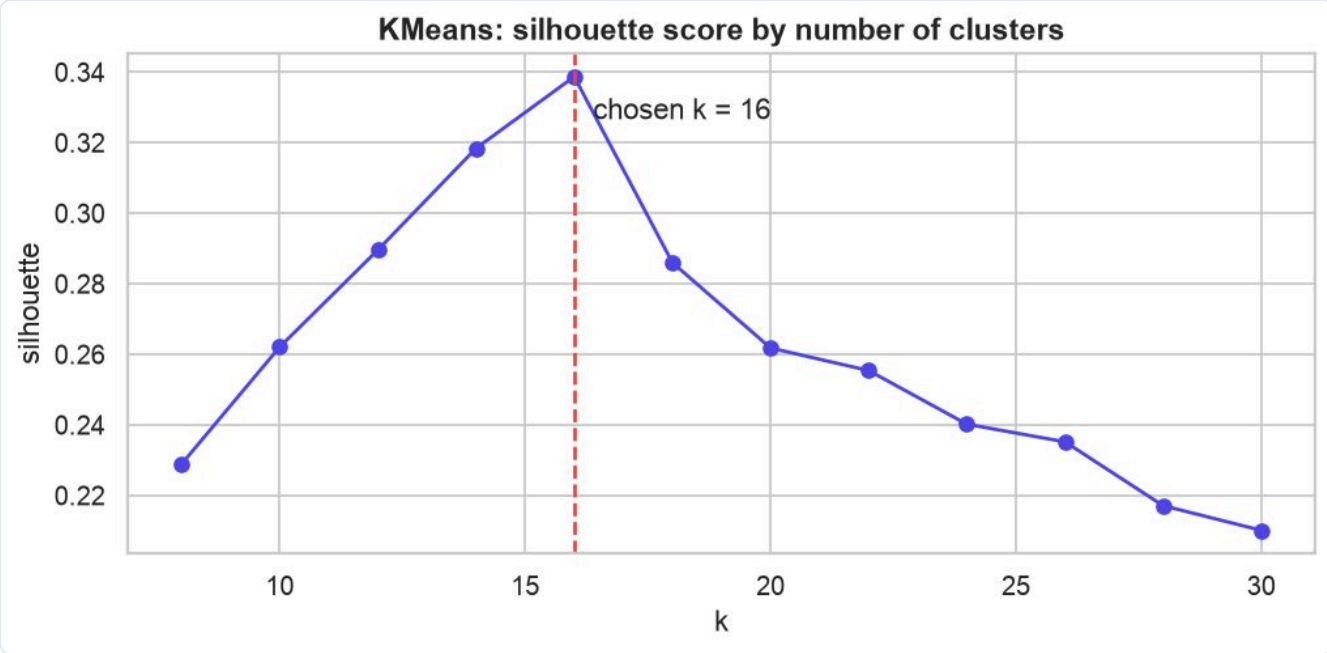
TF-IDF on titles and descriptions (with a light genre signal), reduced to 150 SVD "topic" dimensions, falls into **16 clear clusters** (silhouette peaks at k = 16, 0.339). Each one reads as a recognisable shelf:

- **Mystery & Thriller:** Poirot, *The Girl on the Train*, Jack Reacher.
- **Money & Finance:** *Think and Grow Rich*, *Rich Dad Poor Dad*, *The Intelligent Investor*.
- **Science & Technology:** *A Brief History of Time*, *Astrophysics for People in a Hurry*.
- **Spirituality:** *The Power of Your Subconscious Mind*, *The Four Agreements*.

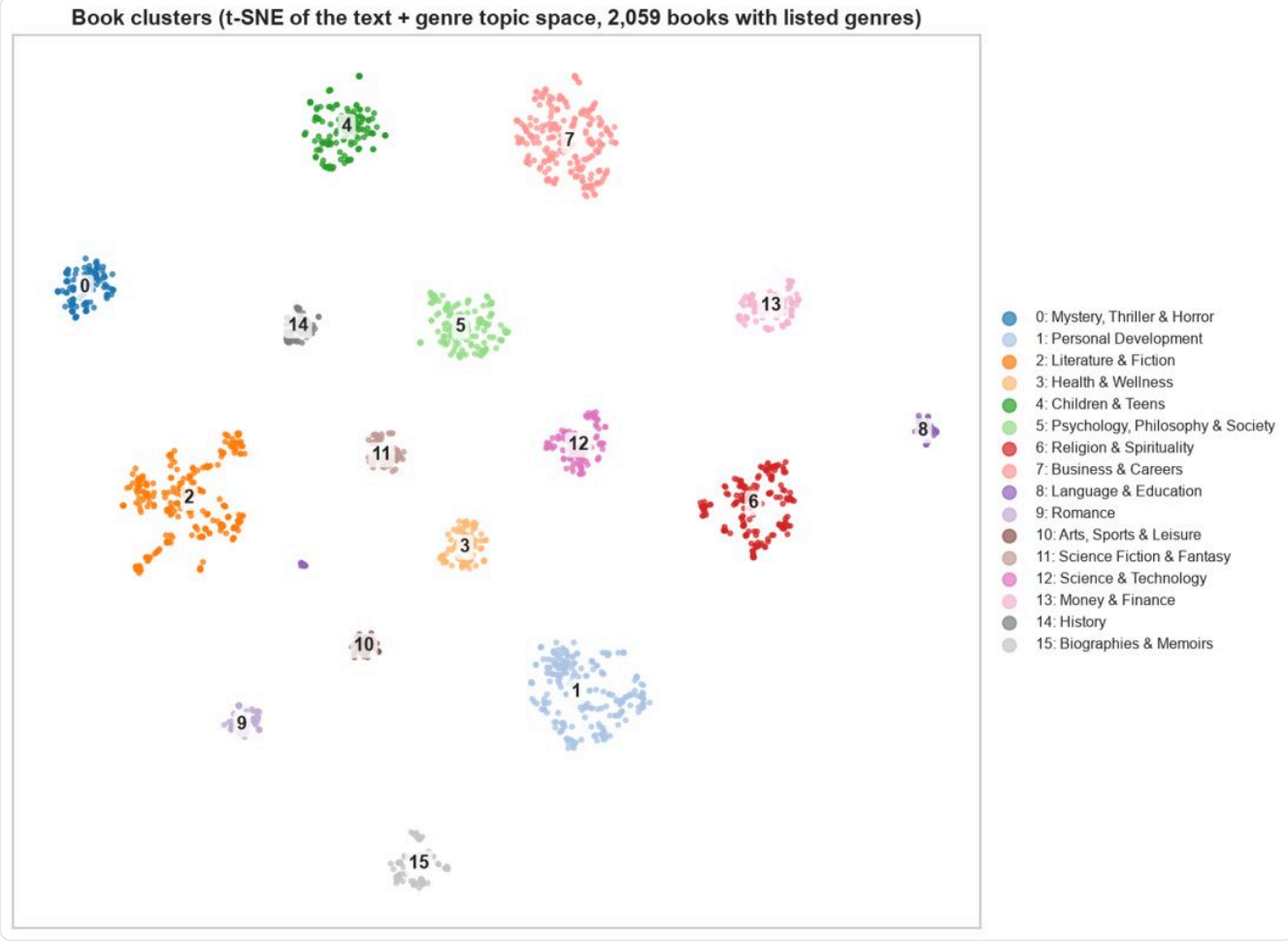
DBSCAN as a cross-check. DBSCAN is very sensitive to its radius (eps). At eps 0.3 it calls 97% of books noise; at 1.0 one cluster swallows 87%. At eps 0.8–0.9 it finds 16–18 clusters with almost no noise, and they **agree with KMeans at an adjusted Rand index of 0.98**. Two different algorithms find the same structure, so the clusters are real rather than an artefact of KMeans. KMeans is kept because it gives every book a cluster, and its centroids can place the 3,997 books without a listed genre.

Most similar pairs by description alone (different authors, same cluster):

- the same story in two languages (*The Blue Jackal* / *Rangeela Geedad*);
- BBC radio dramatisations (*Good Omens* / *Neverwhere*);
- CliffsNotes study guides;
- *Good Night Stories for Rebel Girls* volumes 1 and 2;
- consecutive Malazan novels.



Silhouette score by k



t-SNE map of the 16 clusters

#	Cluster	Books	Fitted	Examples
1	Personal Development · life, success, people	633	237	How to Win Friends and Influence People · The Subtle Art of Not Giving a F*ck: A Counterintuitive Approach to Living a Good Life
7	Business & Careers · business, leadership, management	530	242	Thinking, Fast and Slow · The Art of War
0	Mystery, Thriller & Horror · murder, thriller, poirot	508	106	The Girl on the Train · Inferno
6	Religion & Spirituality · spiritual, god, meditation	508	183	The Power of Your Subconscious Mind · The Four Agreements
2	Literature & Fiction · novel, love, story	470	280	The Nightingale · The Alchemist: A Fable About Following Your Dream
4	Children & Teens · stories, kid, diary	441	154	The Fault in Our Stars · Mockingjay: The Hunger Games, Book 3
11	Science Fiction & Fantasy · dark, doctor, blood	385	70	The Martian · Outlander: Outlander, Book 1
12	Science & Technology · physics, nature, universe	351	109	Astrophysics for People in a Hurry · A Brief History of Time: From Big Bang to Black Holes
9	Romance · love, dirty, love tinder	341	62	Me Before You · Fifty Shades Freed: Book Three of the Fifty Shades Trilogy
3	Health & Wellness · body, weight, health	335	92	Wild · When Breath Becomes Air
13	Money & Finance · money, financial, market	309	106	Think and Grow Rich · Rich Dad Poor Dad: What the Rich Teach Their Kids About Money - That the Poor and Middle Class Do Not!
5	Psychology, Philosophy & Society · india, political, world	286	153	Sapiens · Man's Search for Meaning
15	Biographies & Memoirs · autobiography, man, journey	255	95	Becoming · Can't Hurt Me: Master Your Mind and Defy the Odds
14	History · history, india, east	248	63	Endurance: Shackleton's Incredible Voyage · The Prince
8	Language & Education · learn, english, language	235	60	China Rich Girlfriend · Everything Is F*cked: A Book About Hope
10	Arts, Sports & Leisure · music, story, voyage	221	47	Salt, Fat, Acid, Heat: Mastering the Elements of Good Cooking · Born to Run: The Hidden Tribe, the Ultra-Runners, and the Greatest Race the World Has Never Seen

DBSCAN comparison

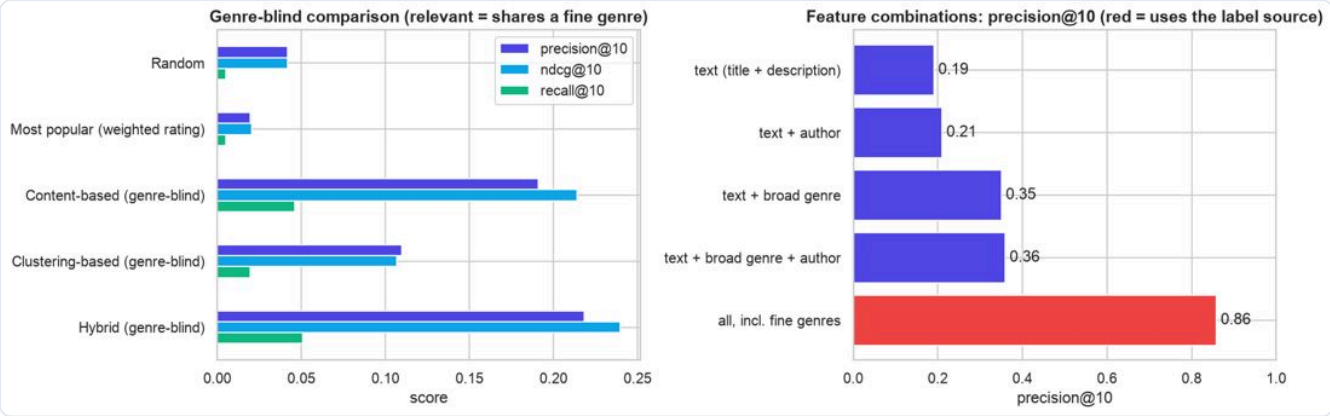
eps	Clusters	Noise	Largest cluster	Silhouette	Agreement with KMeans (ARI)
0.3	10	97.0%	0.4%	0.800	-0.00
0.4	26	88.7%	1.9%	0.600	-0.00
0.5	47	72.4%	3.7%	0.392	0.02
0.6	56	37.1%	8.9%	0.238	0.21
0.7	23	8.3%	12.4%	0.267	0.84
0.8	**18**	**0.9%**	**13.6%**	**0.336**	**0.98**
0.9	16	0.0%	13.6%	0.332	0.98
1.0	5	0.0%	86.6%	0.072	0.05

5. Recommendation models and evaluation

Model	How it scores a candidate
Content-based	Cosine similarity over weighted blocks: text 1.0, broad genre 0.6, fine genres 0.6, author 0.3
Clustering-based	Books in the query's cluster, ranked by weighted rating (title-only cluster members down-weighted)
Hybrid	0.90 × similarity + 0.05 × same cluster + 0.05 × weighted rating; weights grid-searched on validation queries

> **Evaluation design.** The data has no per-user ratings, so a recommendation counts as **relevant** when it shares at least one of the query book's 876 fine Audible genres (**strictly relevant**: two). Test set: 596 books with listed genres, top 10 recommendations each, seed 42 (tuning used a different seed). **Genre-blind** models use only titles and descriptions, so they cannot see the labels; they are the fair comparison. Full-feature (production) models also use the fine genres and are an optimistic upper bound.

Model	Precision@10	Recall@10	nDCG@10	Strict P@10	Avg rating of recs	Coverage
Random	0.042	0.005	0.042	0.006	4.46	95%
Most popular (weighted rating)	0.020	0.005	0.021	0.001	4.89	1%
Content-based (genre-blind)	0.191	0.046	0.214	0.052	4.46	89%
Clustering-based (genre-blind)	0.110	0.019	0.107	0.013	4.77	9%
Hybrid (genre-blind)	**0.218**	**0.051**	**0.240**	**0.053**	**4.49**	**88%**
Content-based (full features)	0.859	0.249	0.905	0.385	4.46	86%
Clustering-based (full features)	0.206	0.043	0.204	0.034	4.77	8%
Hybrid (full features)	0.857	0.248	0.904	0.385	4.49	85%



Genre-blind comparison and feature ablation

Is the difference real? Wilcoxon signed-rank test on per-book nDCG@10 for the same test books (paired, non-parametric):

Model A	Model B	A wins	B wins	Ties	p-value
Hybrid (genre-blind)	Content-based (genre-blind)	261	138	197	2.6e-11
Hybrid (genre-blind)	Clustering-based (genre-blind)	368	99	129	1.2e-41
Content-based (genre-blind)	Random	402	57	137	1.1e-60

Q7. How does genre similarity affect book recommendations?

Genre similarity is the strongest single driver of recommendation quality. Here a recommendation counts as *relevant* when it shares at least one of the query's fine Audible genres; precision is measured over the top 10, across 596 test books.

- **Description text alone** (genre-blind) gets **19% precision@10**, about 5× random (4%). So description NLP alone recovers genre-mates.
- **Adding the broad genre** (17 classes) almost doubles precision, from 0.19 to **0.35** (see Q9).
- **Using the fine genres directly** reaches 0.86. That number is optimistic, because the fine genres are also what defines "relevant"; treat it as an upper bound, not a fair score.
- **The cluster-based model** recommends the best-rated books in the query's cluster. Its precision is lower (0.11 genre-blind) and it covers only 9% of the catalogue. But its recommendations rate highest (4.77 average): it trades personalisation for safe, popular picks.
- **The hybrid** (content + same cluster + quality, weights tuned on separate validation queries) beats pure content in the genre-blind setting (0.218 vs 0.191 precision, 0.240 vs 0.214 nDCG) and also recommends better-rated books.

Which features matter? Rating prediction (RMSE)

Features	Precision@10	nDCG@10	Strict P@10
text (title + description)	0.191	0.214	0.052
text + author	0.210	0.239	0.059
text + broad genre	0.350	0.364	0.088
text + broad genre + author	**0.360**	**0.382**	**0.092**
all, incl. fine genres	0.859	0.905	0.385

Rating predictor	RMSE	MAE
Global mean (baseline)	0.2743	0.2021
Genre mean	0.2694	0.1966
Author mean	0.2542	0.1861
k-NN content (text only)	0.2686	0.1968
k-NN content (all features)	0.2571	0.1864
Hybrid (k-NN + author mean)	**0.2512**	**0.1830**

Q9. Which combination of features provides the most accurate recommendations?

Feature blocks were added one at a time to the content-based model:

features	precision@10	nDCG@10
text (title + description)	0.191	0.214
+ author	0.210	0.239
+ broad genre	0.350	0.364
+ broad genre + author	**0.360**	**0.382**
all, incl. fine genres (upper bound)	0.859	0.905

Text + broad genre + author is the best fair combination. Every block adds something, and the genre block adds by far the most. The production model also uses the fine genres, because they are real information about every book that has them.

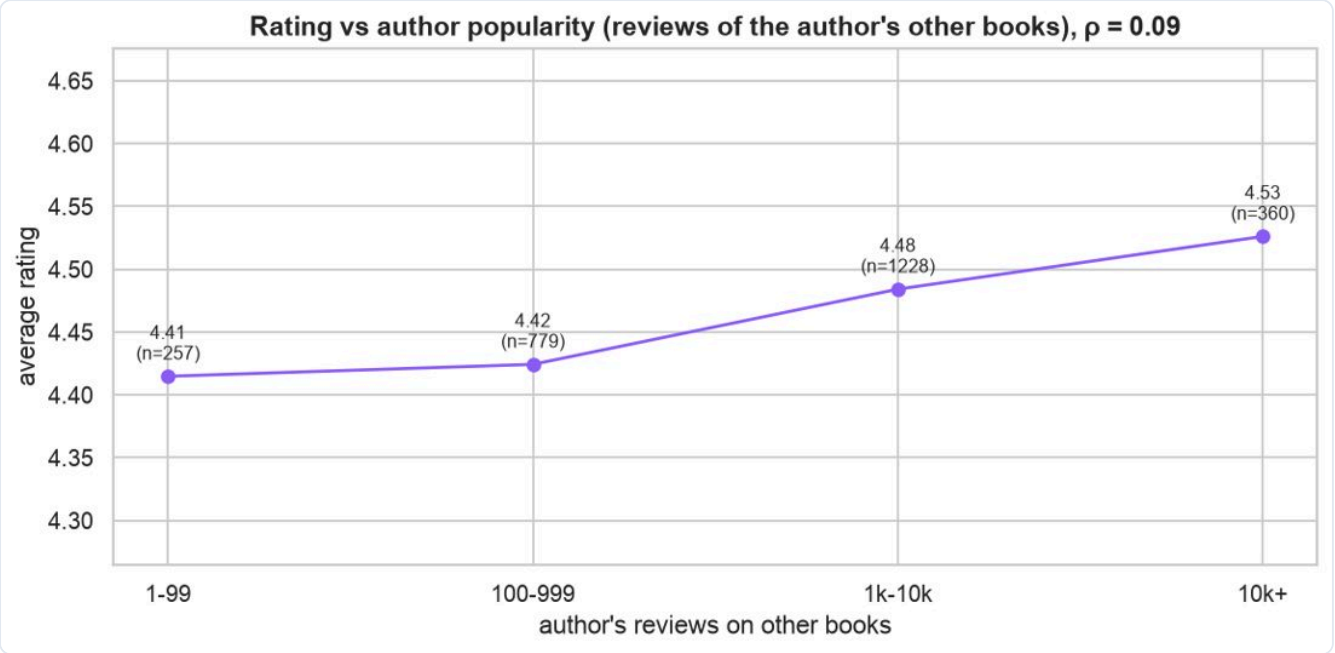
For **rating prediction** (RMSE, 1,045 held-out books), averaging the k-NN content prediction with the author's mean rating does best: **RMSE 0.251 vs 0.274** for always predicting the mean (–8%). That gain is small because ratings barely vary (std 0.35). Ratings are a weak signal in this data, so the system ranks by similarity first and uses quality as a tie-breaker.

Q8. What is the effect of author popularity on book ratings?

Here, author popularity means the reviews on the author's *other* books, so a single blockbuster is not compared with itself. There's a small, consistent positive effect:

- Books by authors whose other work has 10k+ reviews average **4.53**, against **4.42** for authors with under 100.
- The share of books rated below 4.0 drops from 8.2% to 2.8%.

The correlation is weak ($\rho = 0.09$, $p \approx 10^{-6}$): **an established author is a safer bet, not a guarantee.** This is why the author is a small feature block (weight 0.3) in the content model, and why "Author mean" is a strong rating predictor (Q9, RMSE table).



Q8. Rating vs author popularity

6. Scenario questions

S1. A new user likes science fiction. Which top 5 books should be recommended?

A new user has no history (cold start), so I take books whose Audible genres include "Science Fiction" (children's titles excluded) and rank them by weighted rating:

1. *Morning Star* (Pierce Brown)
2. *So Long, Lollipops* (Sarah Lyons Fleming)
3. *The Martian* (Andy Weir)
4. *Jurassic Park* (Michael Crichton)
5. *Thrawn* (Timothy Zahn)

All rate 4.7–4.8, and *The Martian* has 33k reviews. Once they've started, the hybrid model suggests *Red Rising*, *The Restaurant at the End of the Universe*, *Blade Runner*, *Hyperion* and *Anthem*.

S2. A user rated thrillers highly. Recommend similar books.

The user's history is five popular thrillers they rated highly:

- *Then She Was Gone*
- *The Hit*
- *Tripwire*
- *Verity*
- *The Visitor*

The hybrid model averages their feature vectors into one profile. It returns ten books, **all Mystery, Thriller & Horror**, mixing:

- series continuations (*Killing Floor*, *Nothing to Lose*);
- authors they already like (*Walk the Wire* by Baldacci);
- new authors (*Moonflower Murders*, *Rebecca*, *The Day of the Jackal*);
- Indian thrillers (*Chanakya's Chant*, *Private Delhi*), which suit an Audible India audience.

S3. Identify books that are highly rated but have low popularity (hidden gems)

A hidden gem here means:

- a rating of at least **4.5**;
- review counts in the **bottom quarter** of the catalogue (65 or fewer);
- but at least 5 reviews, so the rating isn't one fan.

That gives **504 gems**. Ranking them by weighted rating rewards gems with slightly more evidence. The top of the list:

- *Raven's Hart* (5.0, 37 reviews)
- *Girls' Club* (4.8, 61)
- *Grey Sunshine: Stories from Teach for India* (4.8, 55)
- *Move* (4.8, 50)
- Osho's *A Course in Meditation* (4.9, 30)

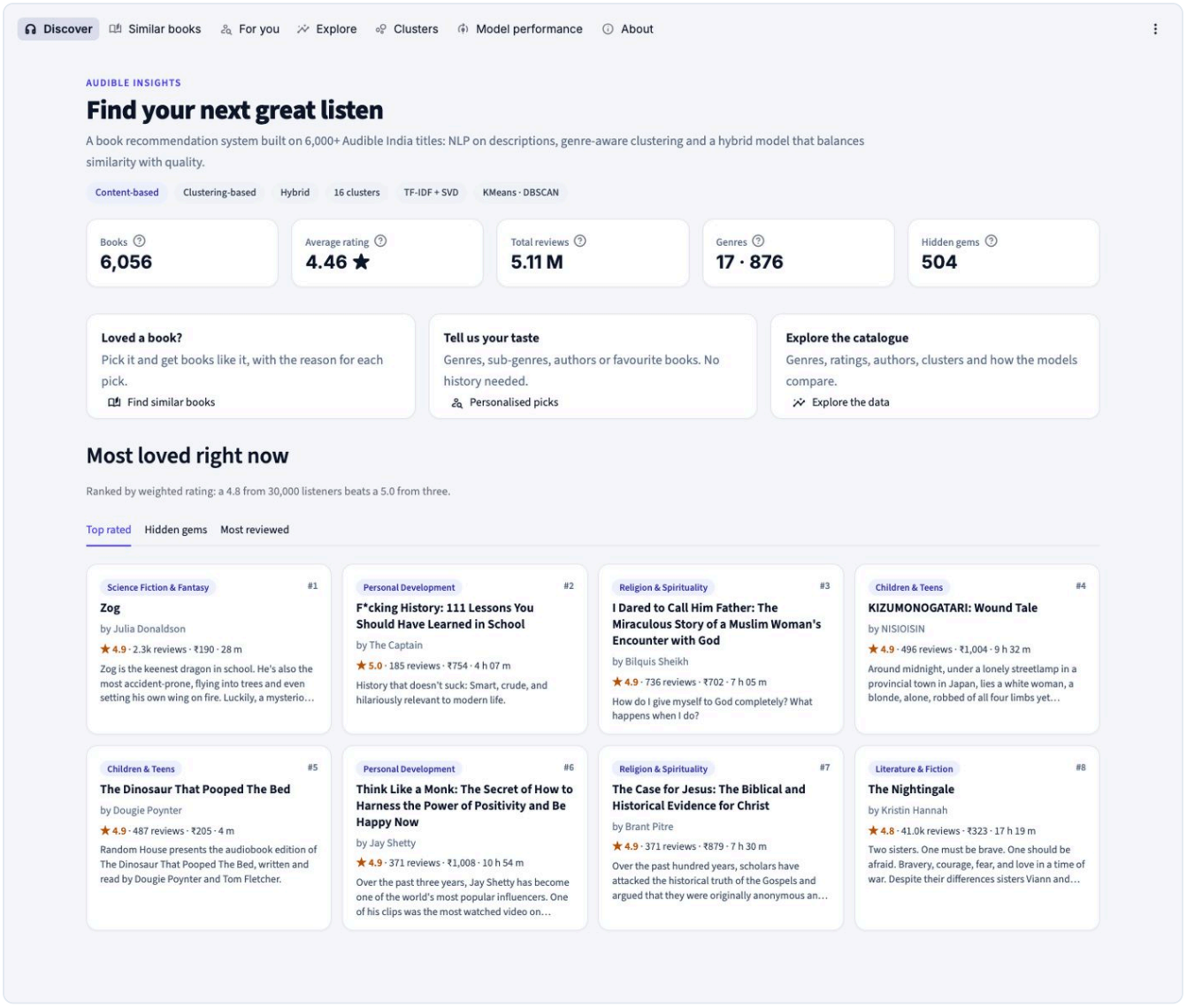
The app has a "hidden gems only" filter, so a user can find them in any genre.

7. Streamlit application

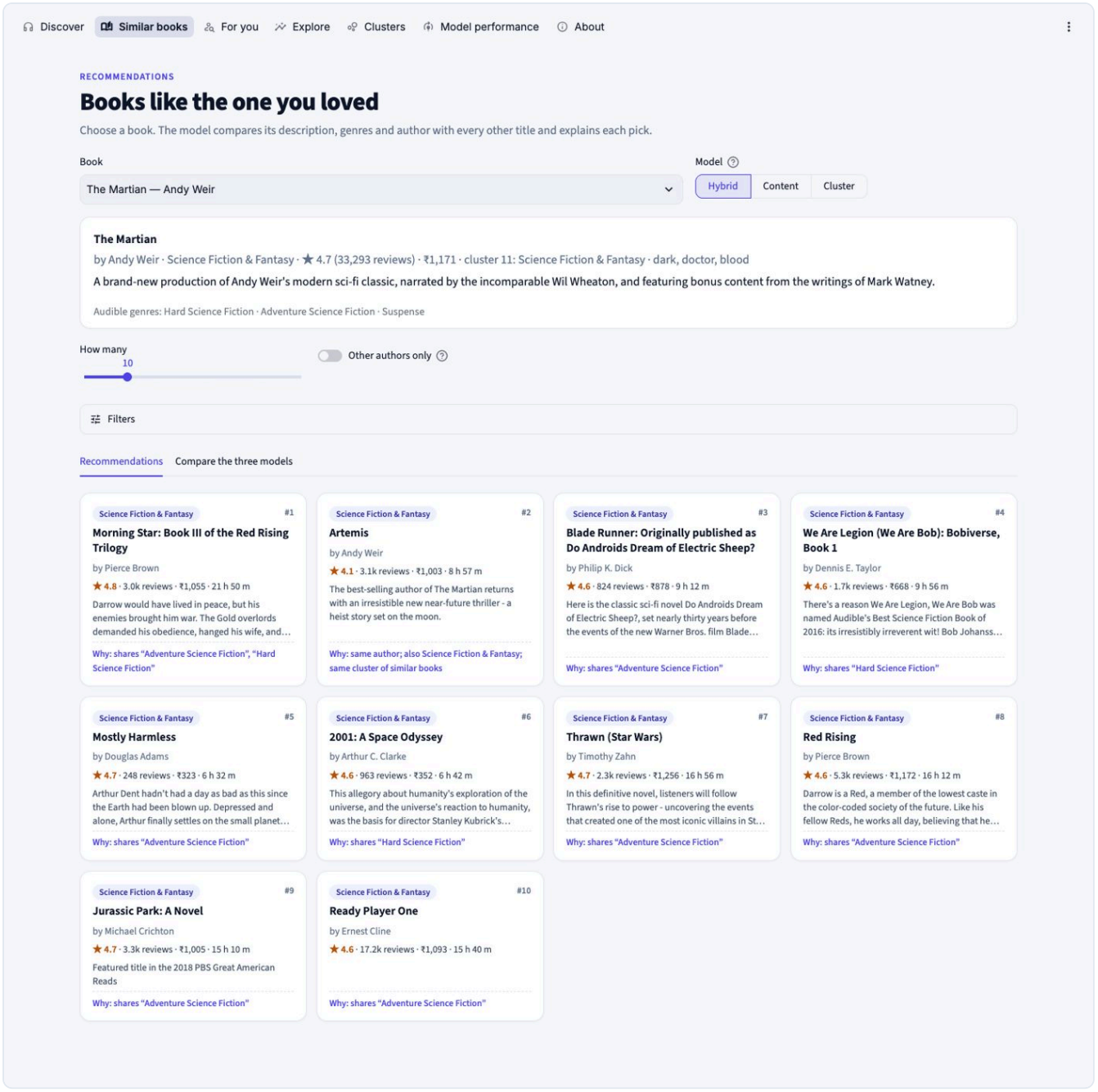
Seven pages:

- **Discover:** KPIs plus top-rated, hidden-gem and most-reviewed shelves.
- **Similar books:** pick a book and a model, get explained recommendations, and compare the three models side by side.
- **For you:** genres, sub-genres, authors and liked books, with quick-start scenarios.
- **Explore:** the EDA, interactive.
- **Clusters:** map, cluster browser, KMeans vs DBSCAN.
- **Model performance.**
- **About.**

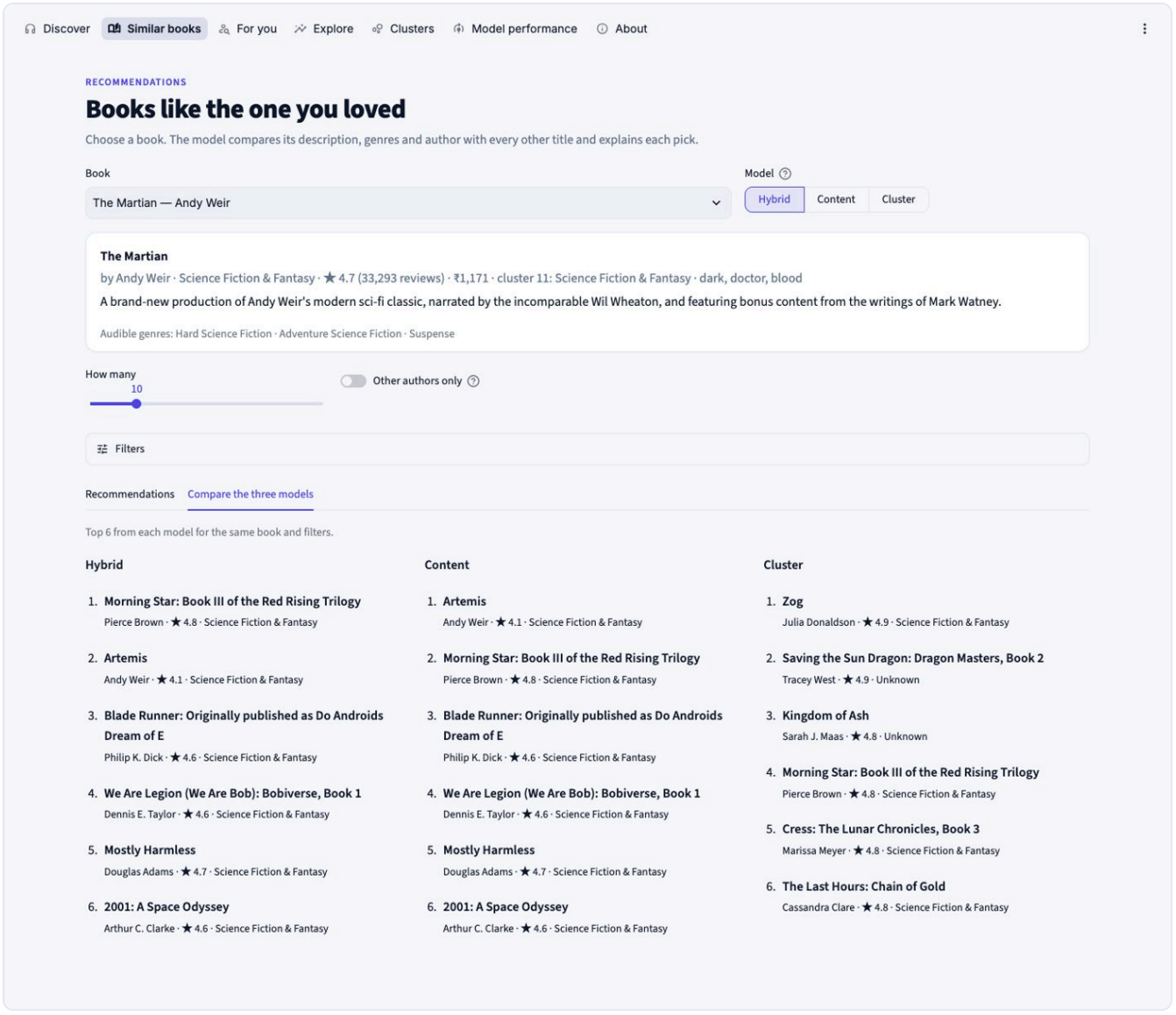
Every recommendation page has filters: genre, minimum rating and reviews, price, length, unrated books and hidden gems.



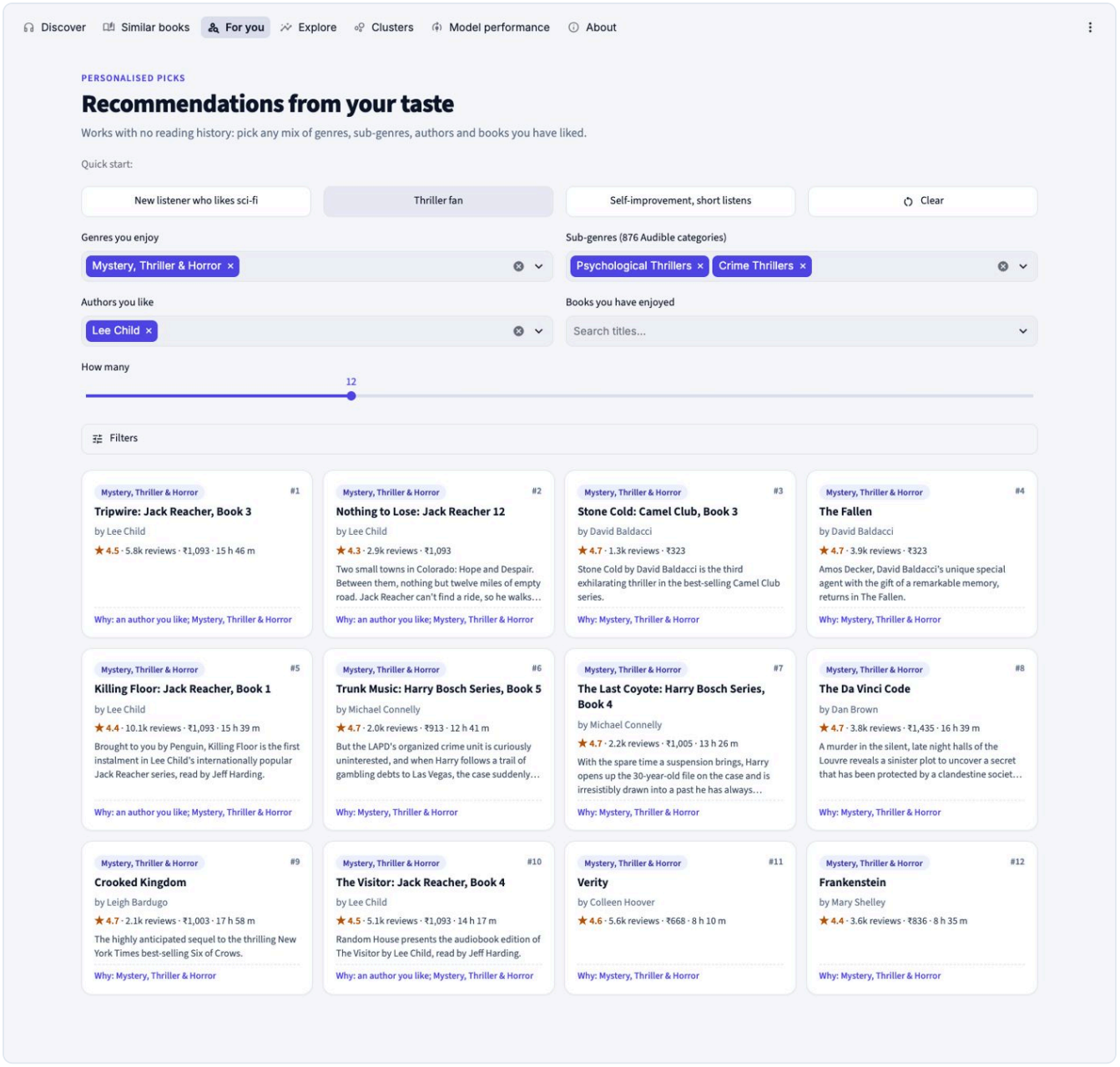
Discover: KPIs and shelves



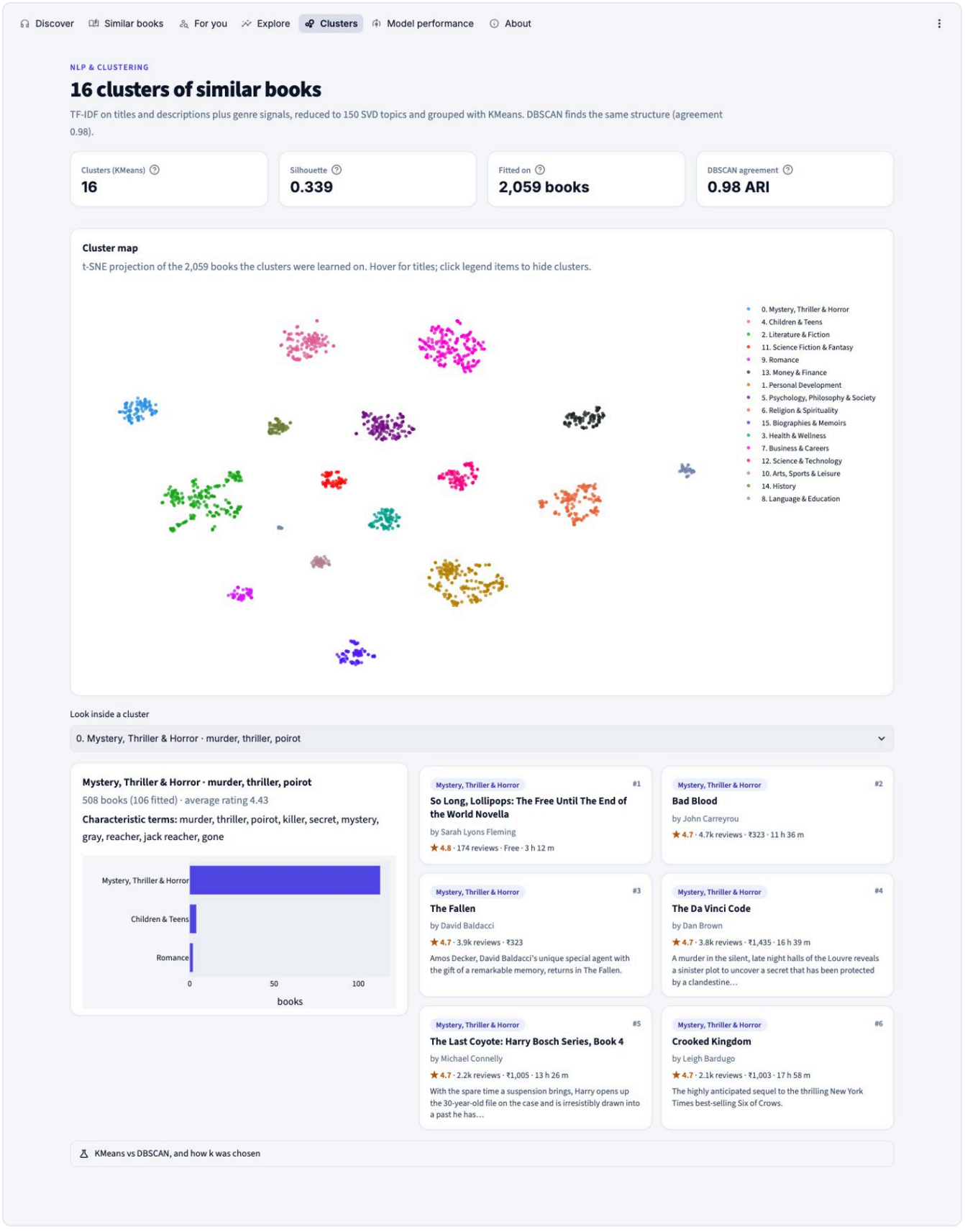
Similar books: The Martian, hybrid model, with reasons



The three models side by side for the same book



For you: the thriller-fan quick start



Clusters page

Discover | Similar books | For you | Explore | Clusters | **Model performance** | About

EVALUATION

How good are the recommendations?

No per-user ratings exist, so a recommendation counts as relevant when it shares at least one of the query book's 876 fine Audible genres. Scores are averaged over 596 test books, top 10 recommendations each.

Hybrid precision@10	Hybrid nDCG@10	Best fair features	Rating RMSE
0.218	**0.240**	**0.360**	**0.251**
↑ 5.2× random	↑ +12% vs content		↓ -8% vs mean

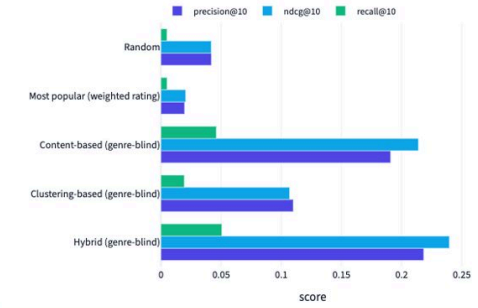
Model comparison

Genre-blind models use only titles and descriptions, so they cannot see the labels being scored. That makes them the fair comparison. **Full-feature** models (production) also use the fine genres, which define relevance, so their scores are an optimistic upper bound.

model	precision@10	recall@10	ndcg@10	strict_precision@10	avg_rating_of_recs	catalogue_coverage	uses fine genres
Random	0.042	0.005	0.042	0.006	4.46	95%	☐
Most popular (weighted rating)	0.020	0.005	0.021	0.001	4.89	1%	☐
Content-based (genre-blind)	0.191	0.046	0.214	0.052	4.46	89%	☐
Clustering-based (genre-blind)	0.110	0.019	0.107	0.013	4.77	9%	☐
Hybrid (genre-blind)	0.218	0.051	0.240	0.053	4.49	88%	☐
Content-based (full features)	0.859	0.249	0.905	0.385	4.46	86%	☑
Clustering-based (full features)	0.206	0.043	0.204	0.034	4.77	8%	☑
Hybrid (full features)	0.857	0.248	0.904	0.385	4.49	85%	☑

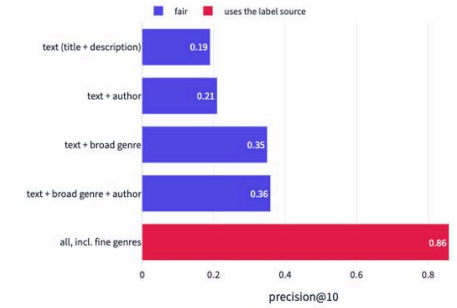
Genre-blind comparison

Hybrid > content > clustering > baselines



Which features matter?

Content-based model, one feature block added at a time



Is the difference real?

Wilcoxon signed-rank test on the per-book nDCG@10 of the same test books (paired, non-parametric). A small p-value means model A beats model B for real, not by chance.

model_a	model_b	metric	mean_a	mean_b	a_better_queries	b_better_queries	ties	wilcoxon_statistic	p_value
Hybrid (genre-blind)	Content-based (genre-blind)	ndcg@10	0.240	0.214	261	138	197	55,037	2.6e-11
Hybrid (genre-blind)	Clustering-based (genre-blind)	ndcg@10	0.240	0.107	368	99	129	93,932	1.2e-41
Content-based (genre-blind)	Random	ndcg@10	0.214	0.042	402	57	137	99,392	1.1e-60

Rating prediction (RMSE)

Predict the rating of 1,045 held-out books (5+ reviews) from similar books. Ratings barely vary (std 0.35), so gains over the mean are modest.

model	rmse	mae
Global mean (baseline)	0.2743	0.2021
Genre mean	0.2694	0.1966
Author mean	0.2542	0.1861
k-NN content (text only)	0.2686	0.1968
k-NN content (all features)	0.2571	0.1864
Hybrid (k-NN + author mean)	0.2512	0.1830

Hybrid weights

Grid-searched on separate validation books: nDCG@10 plus a small reward for well-rated picks.

score = 0.90 × similarity + 0.05 × same cluster + 0.05 × weighted rating

content	cluster	quality	ndcg@10	precision@10	avg_rating_of_recs	objective
0.90	0.05	0.05	0.900	0.844	4.48	0.965
0.80	0.05	0.15	0.896	0.842	4.53	0.965
0.90	0.00	0.10	0.897	0.843	4.51	0.964
0.70	0.20	0.10	0.896	0.842	4.52	0.964
0.80	0.10	0.10	0.896	0.842	4.51	0.964
0.80	0.00	0.20	0.894	0.840	4.55	0.964

Genre inference model (NLP)

Model performance page

8. Deployment

- **Streamlit Community Cloud.** The app runs from the GitHub repo (`app/streamlit_app.py` , pinned `app/requirements.txt`).
- **AWS EC2 + Docker** (`Dockerfile` , `deploy/`):
  - An Amazon Linux 2023 t3.micro runs a start-up script that installs Docker, builds the image and serves the app on port 80.
  - The container restarts on its own, and a health check watches Streamlit.
  - The image is stateless (the model is baked in), so it scales horizontally: an Auto Scaling group behind an Application Load Balancer, or Elastic Beanstalk's Docker platform, as documented in `deploy/AWS_DEPLOY.md` .

9. Insights, limitations and next steps

1. **Data quality was the main problem.** About 2,000 descriptions were scrape failures, and most books have no genre. Treating these honestly as missing, and inferring genre only where confident, mattered more than the choice of model.
2. **Ratings are compressed** (mean 4.46, 61% at 4.5+), so weighted rating is essential. A raw 5.0 is usually a small sample.
3. **Popularity is not quality.** That makes hidden-gem recommendations a real opportunity for libraries and publishers.
4. **Genres are the strongest signal.** Adding the broad genre to description text nearly doubles precision, and the hybrid's quality term keeps recommendations well rated.

- **No user data:** collaborative filtering and true personalisation metrics need per-user ratings.
- **No publication dates:** join them in from Open Library / Google Books to analyse trends.
- **Next:** sentence embeddings (e.g. MiniLM) instead of TF-IDF for short descriptions; re-scrape the blocked pages.

10. Deliverables

Merged and cleaned dataset	`data/processed/books_clean.csv` , `cleaning_report.json`
Scripts	`pipeline.py` , `train.py` , `src/` (cleaning, NLP, recommenders, evaluation, EDA)
EDA notebook	`notebooks/eda.ipynb` : all 12 questions plus my own analysis
Models and metrics	`models/` : recommender, genre classifier, metrics, cluster summary
Streamlit app	`app/`
AWS deployment	`Dockerfile` , `deploy/ec2_user_data.sh` , `deploy/AWS_DEPLOY.md`
Quality assurance	`tests/e2e_app.py` : 163-check browser test of every page and control; the pipeline is byte-for-byte reproducible
Documentation	`README.md` , this report, `reports/LIVE_EVALUATION_GUIDE.md` , `reports/screenshots/`